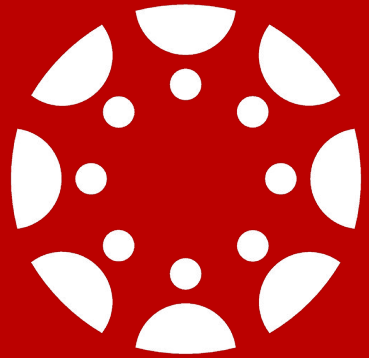
Canvas QuickClick

Team 3:
Caroline Shi, Isabella Webber, Peter Shpelfogel,
Priscilla Park, Dora Chen, Davis Germain



Canvas Quick Click

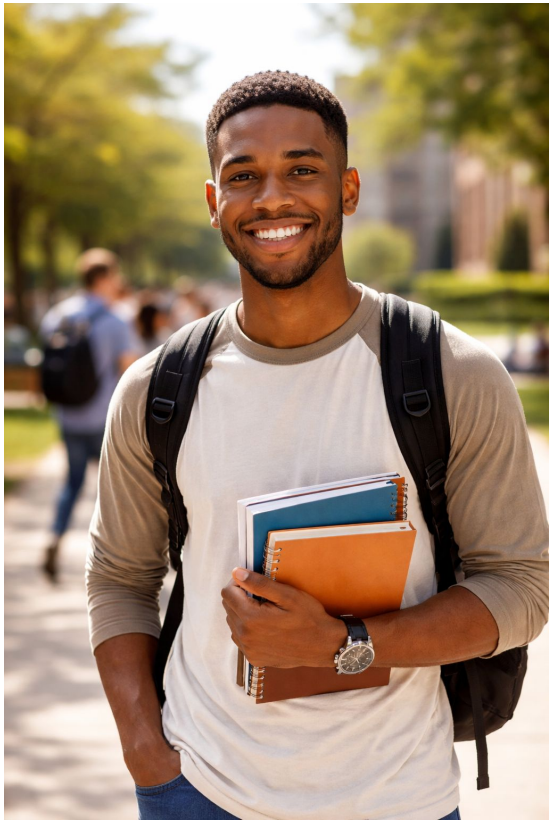
Problem

Solution

Co-creation of Value

Technical Feasibility

Financial feasibility



Name: Derek
Occupation: CMU student

Canvas Quick Click

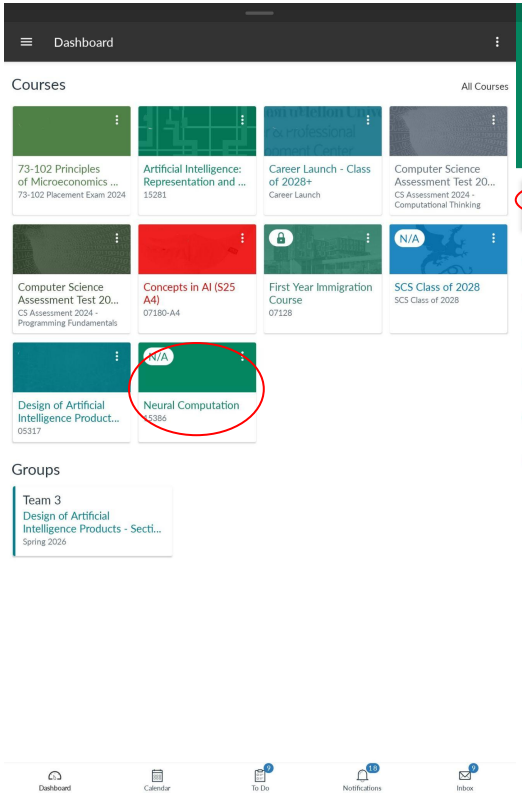
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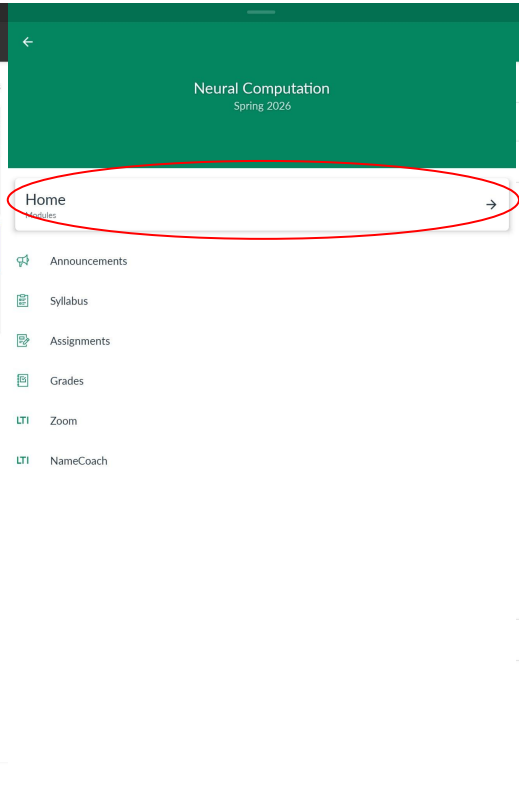
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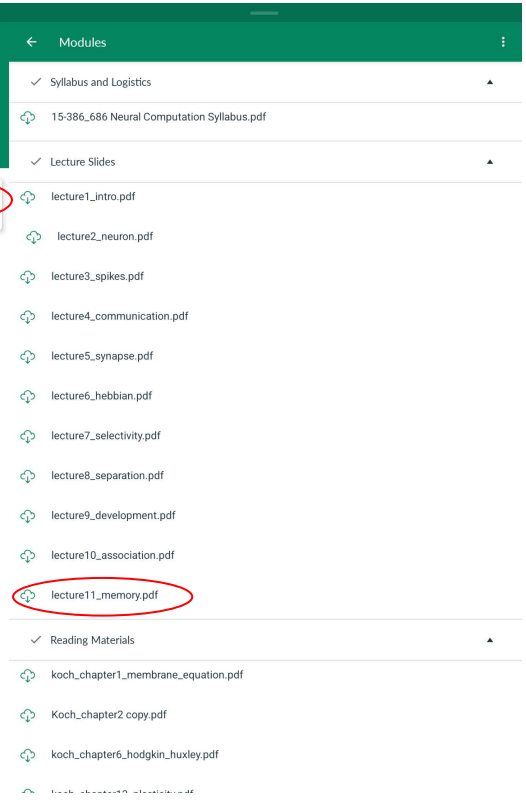
Co-creation of Value

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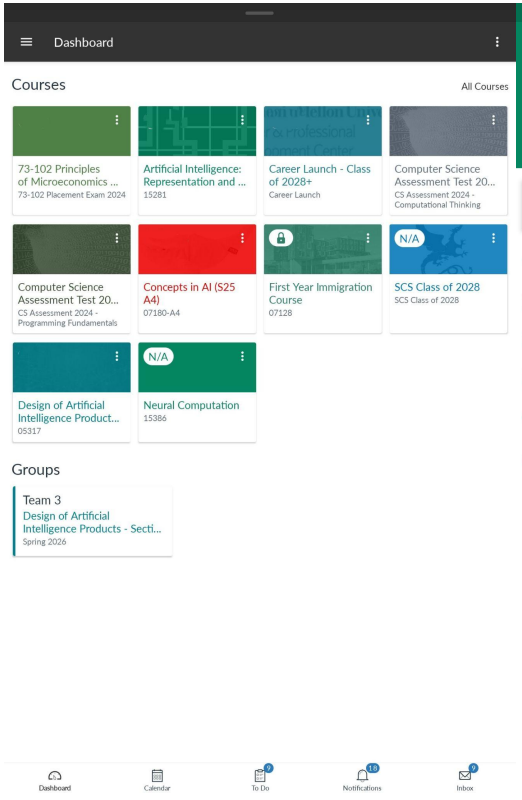
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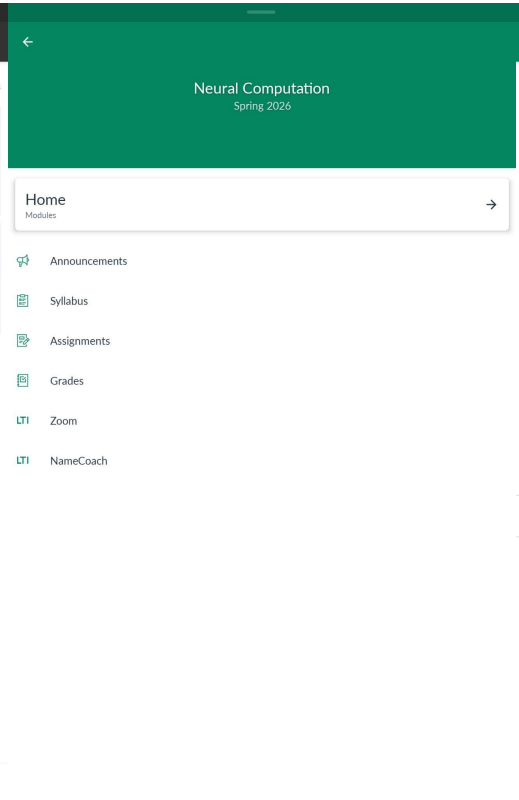
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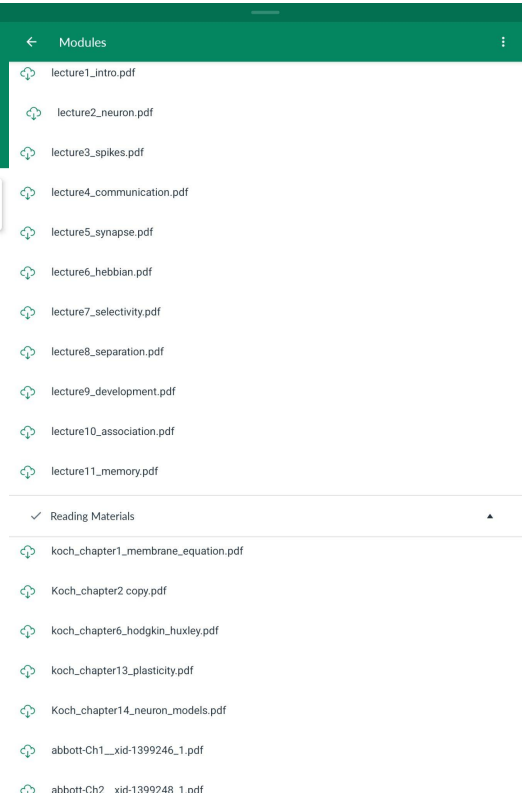
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Canvas Quick Click

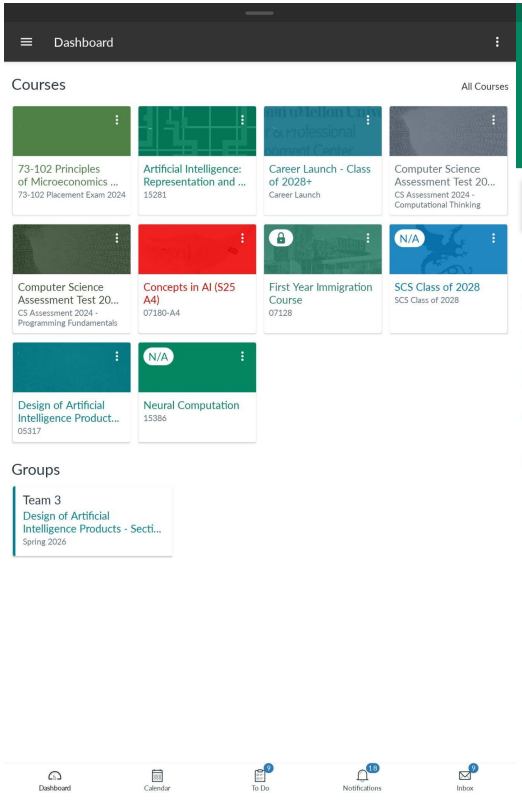
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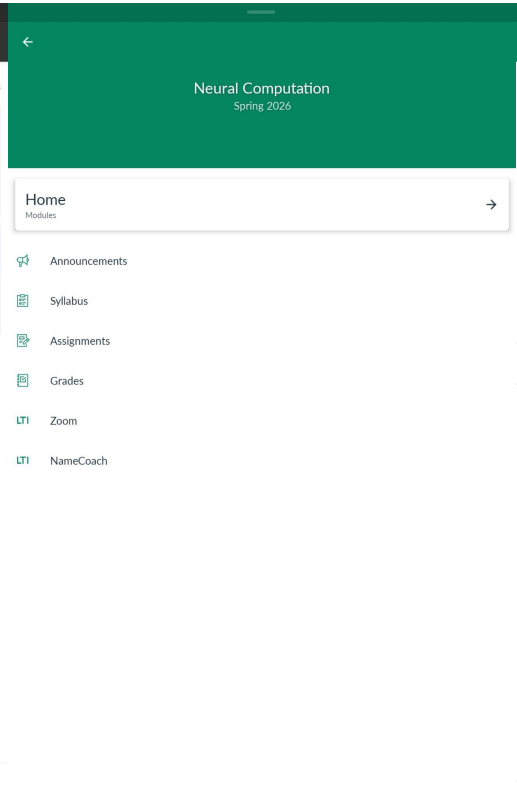
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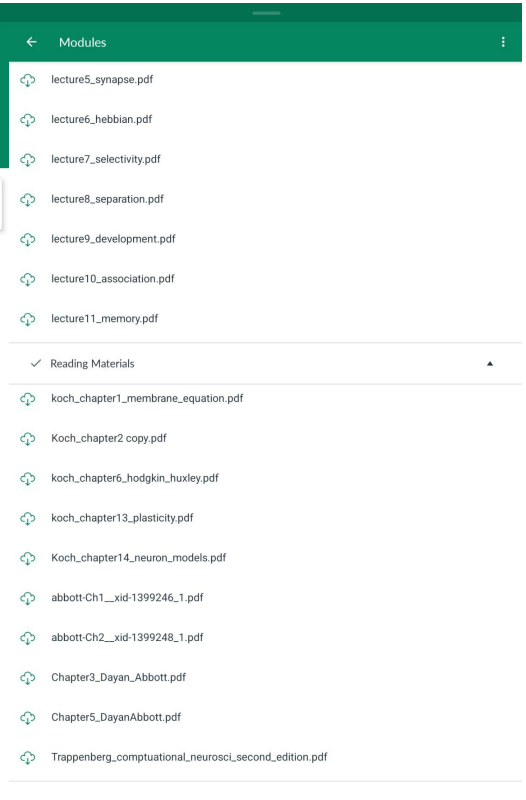
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Canvas Quick Click

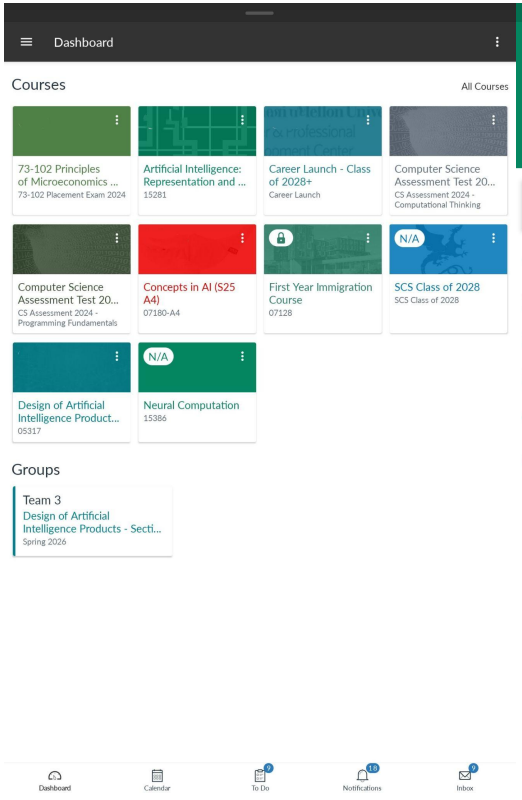
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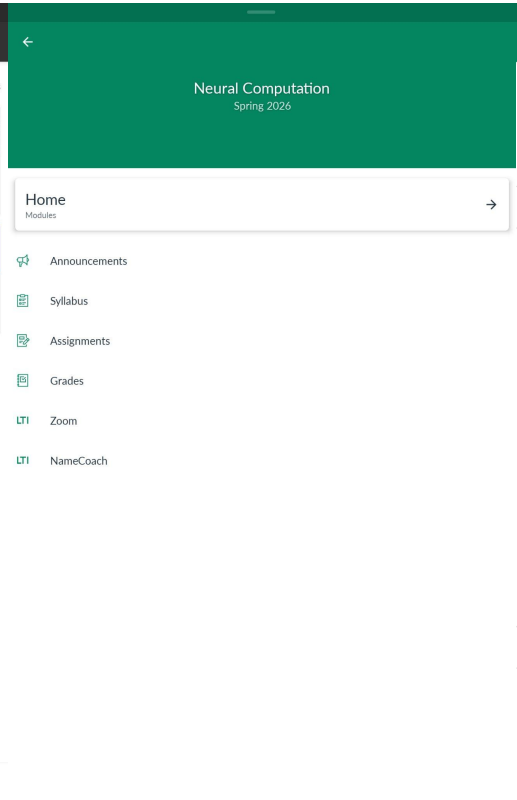
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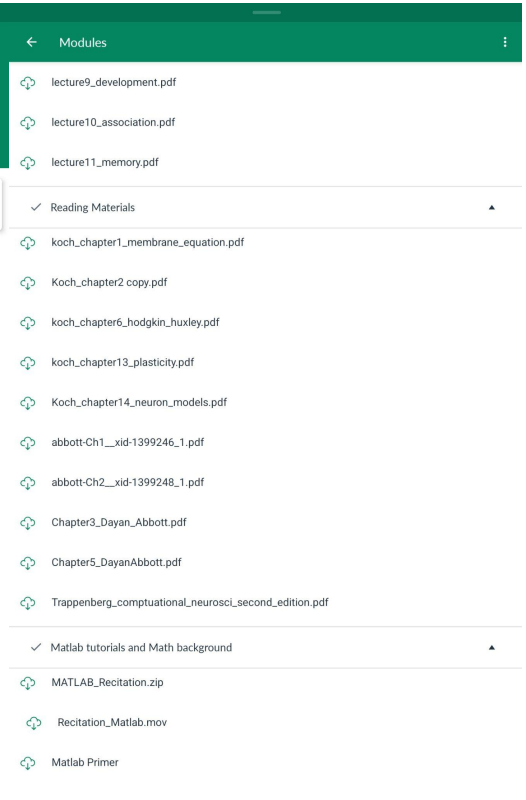
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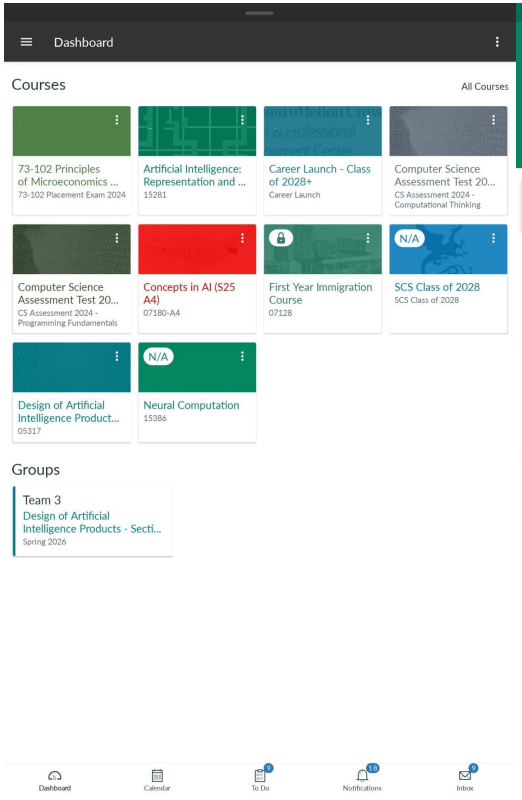
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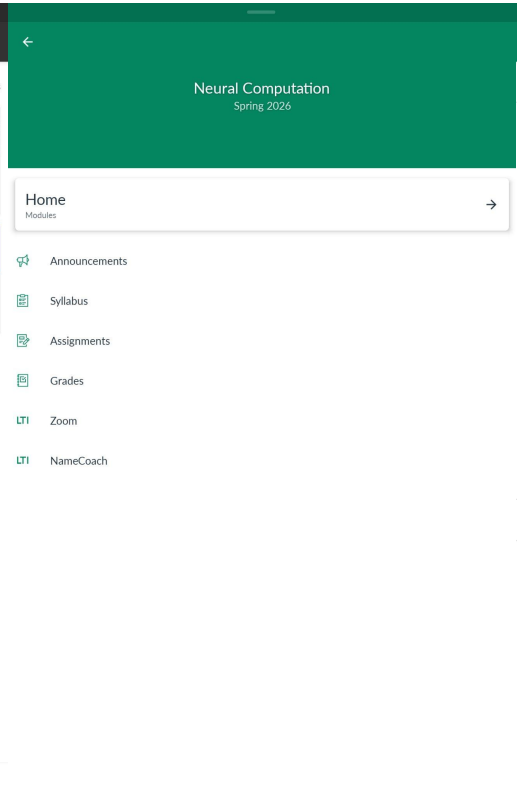
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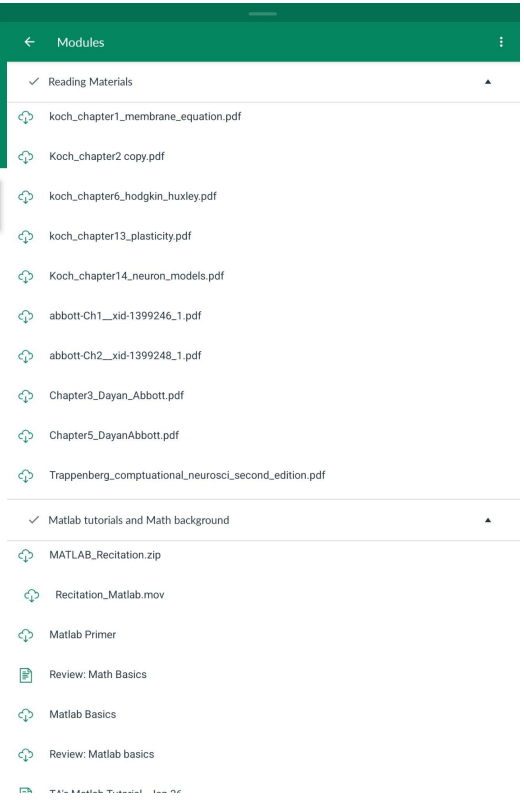
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Canvas Quick Click

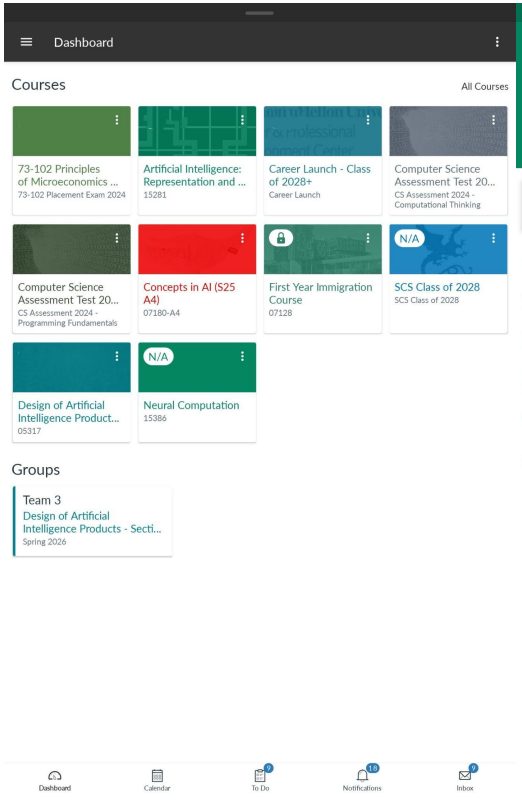
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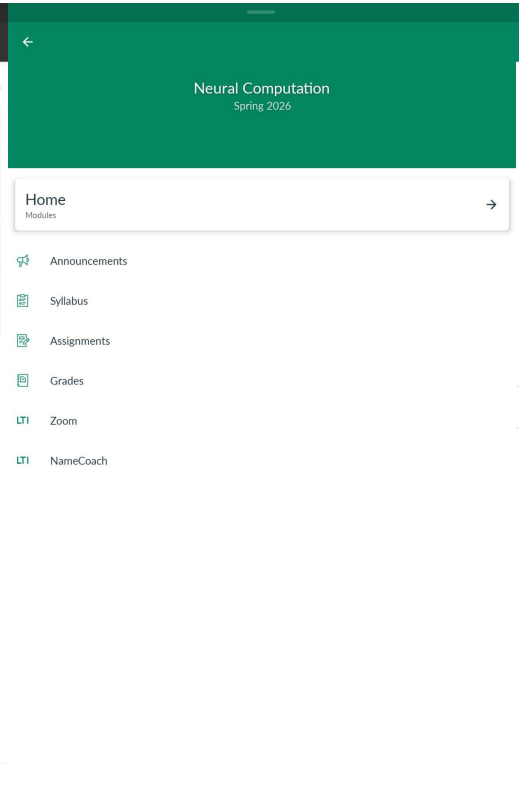
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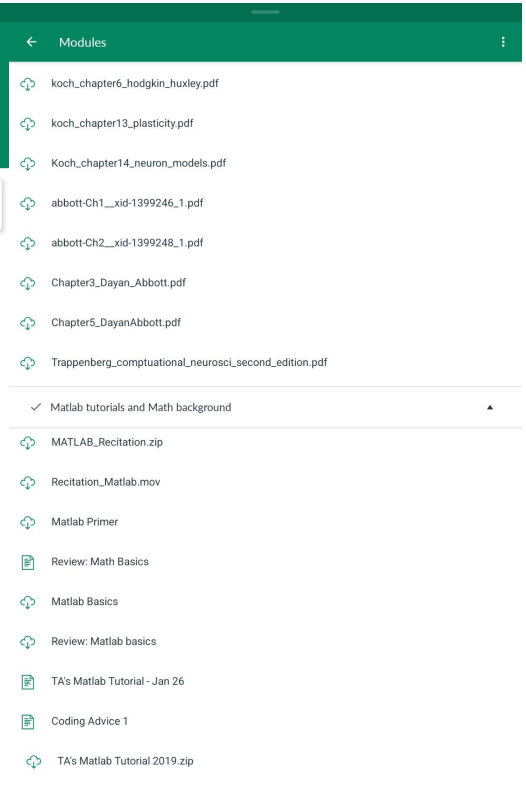
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Canvas Quick Click

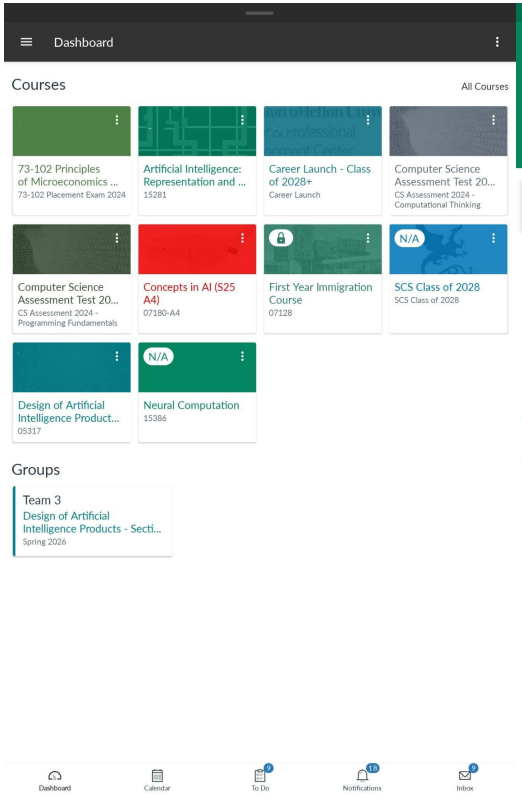
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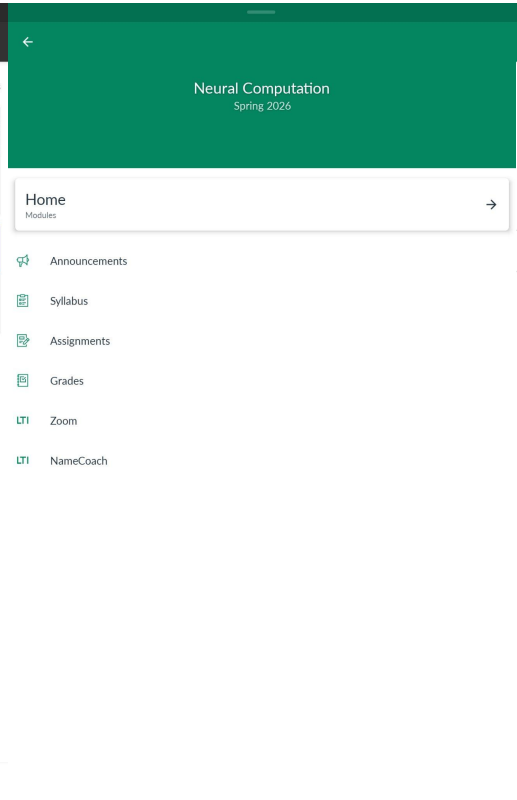
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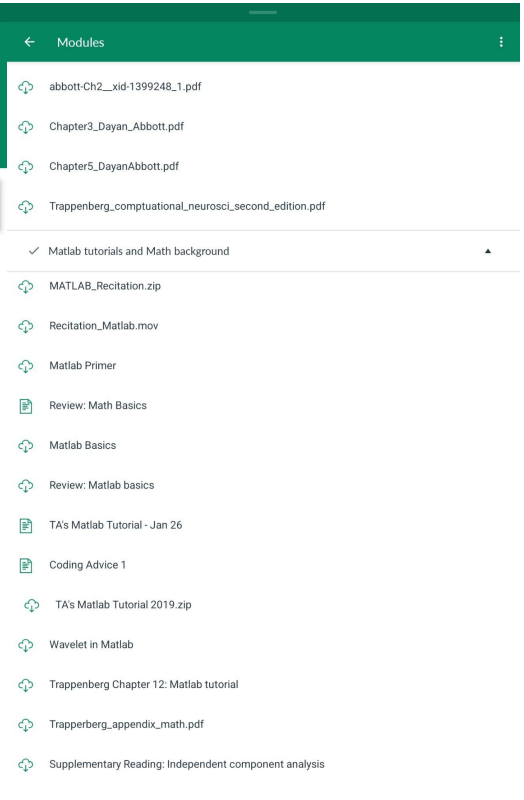
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Canvas Quick Click

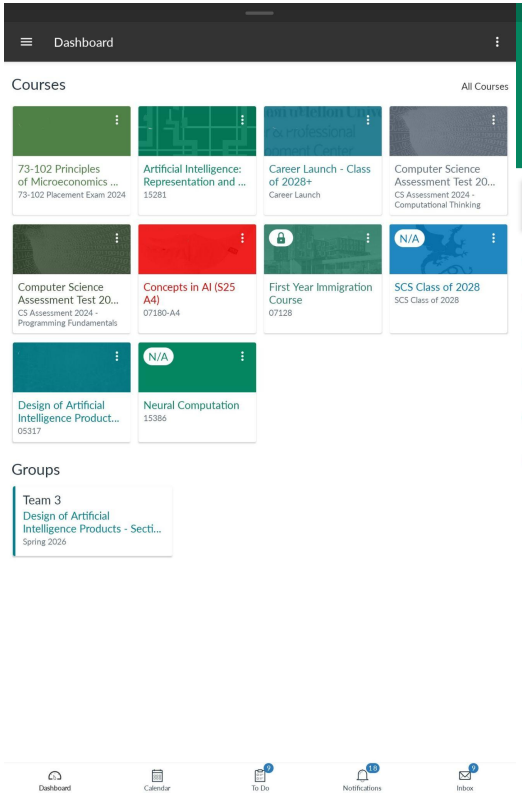
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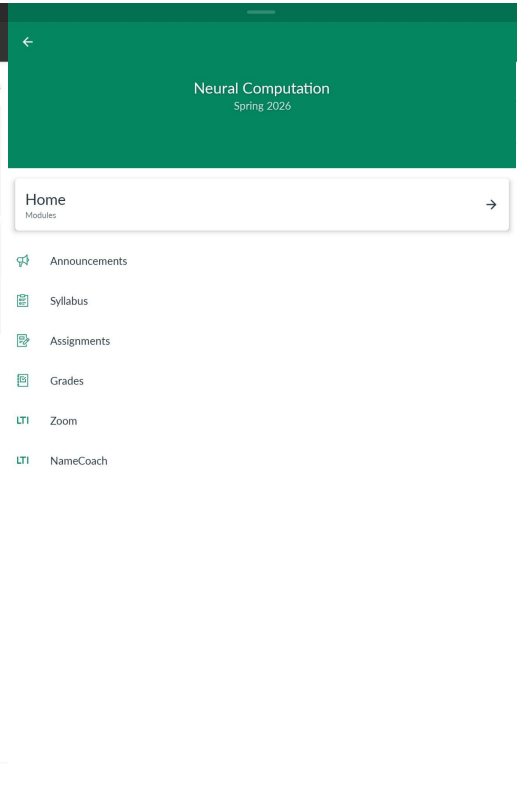
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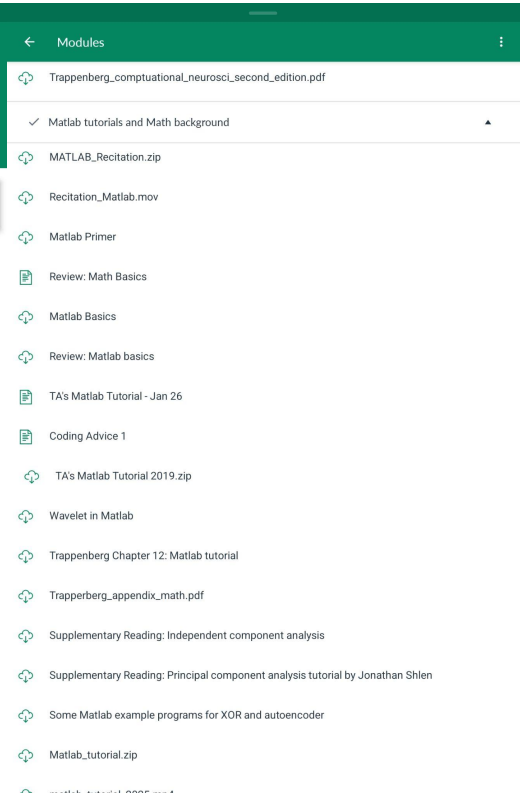
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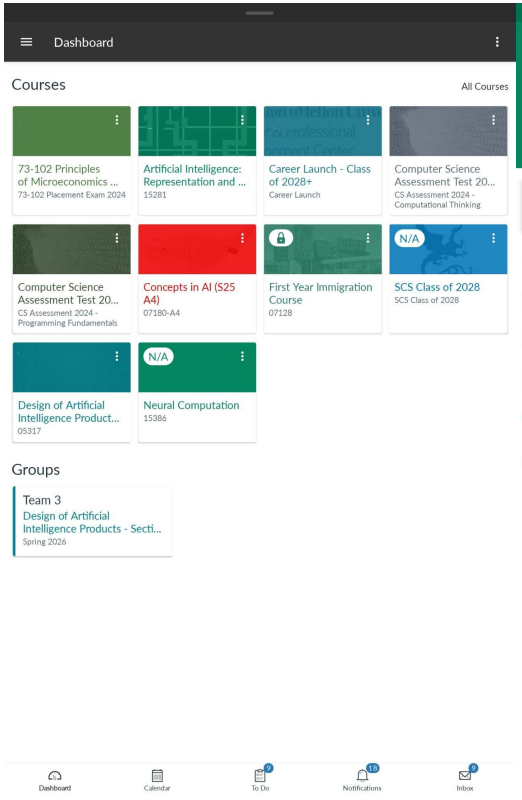
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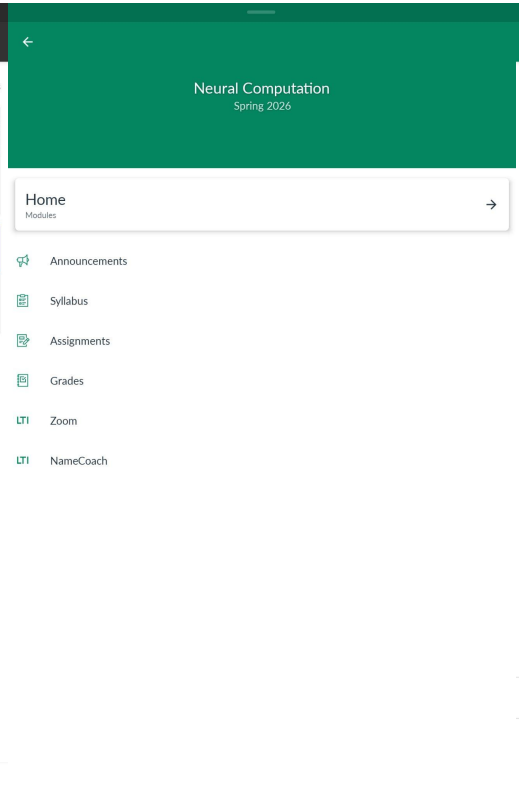
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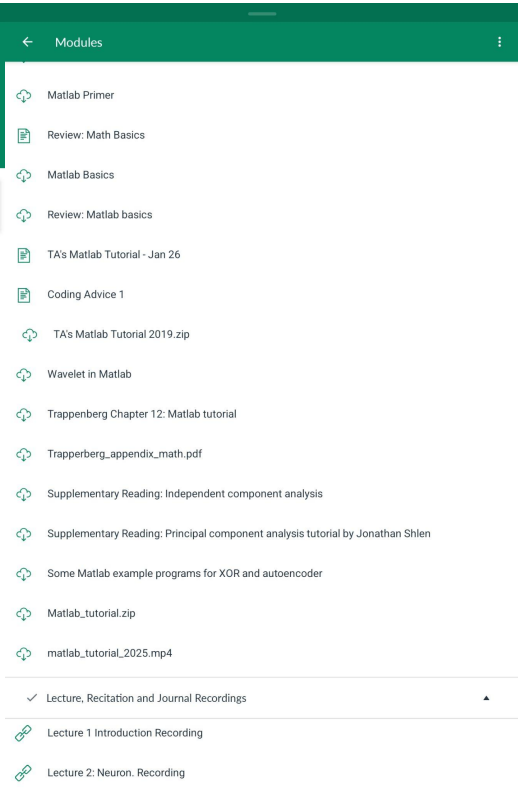
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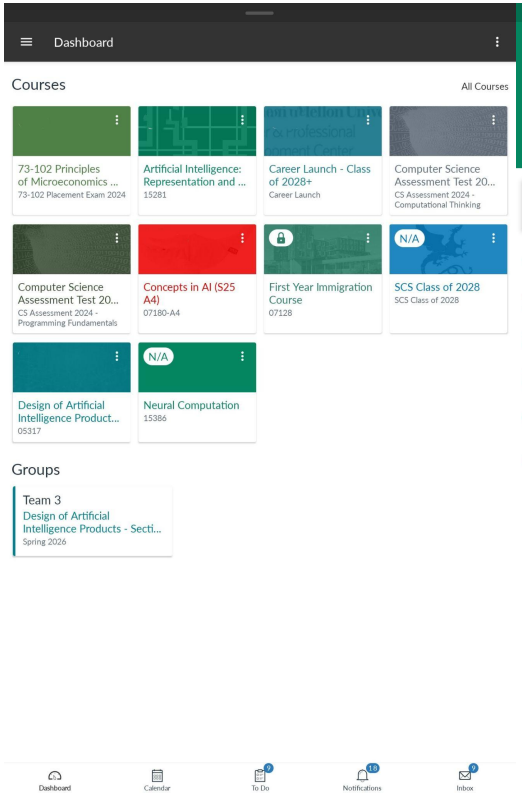
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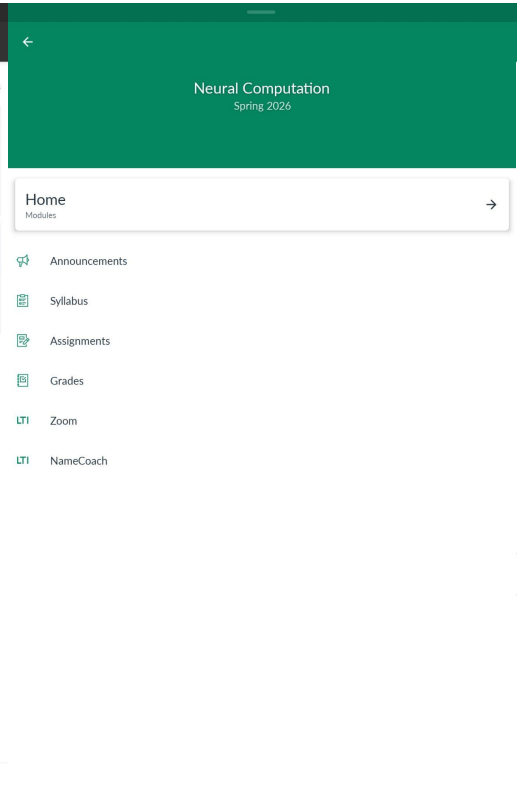
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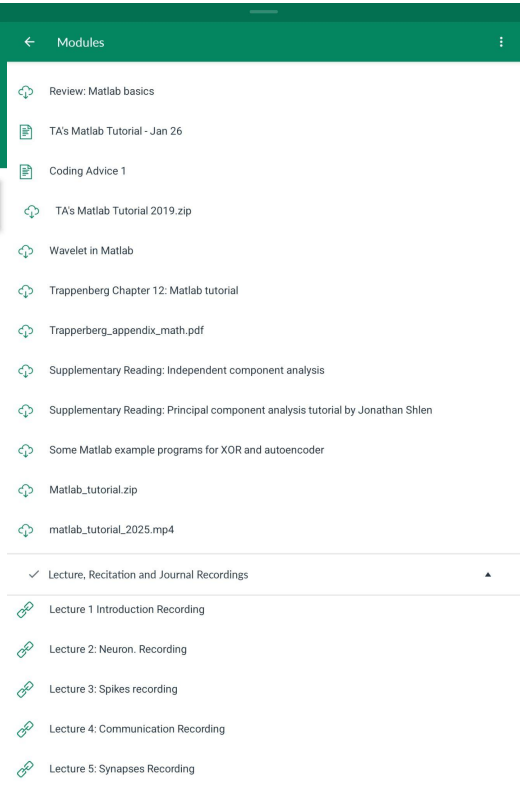
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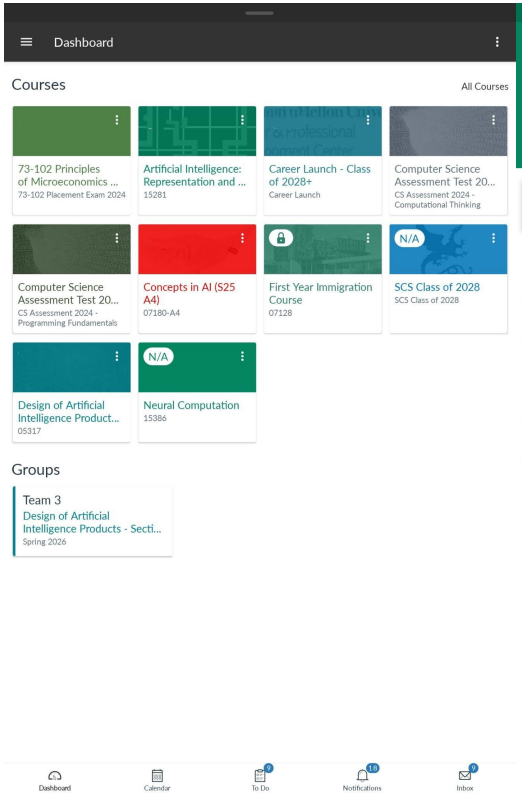
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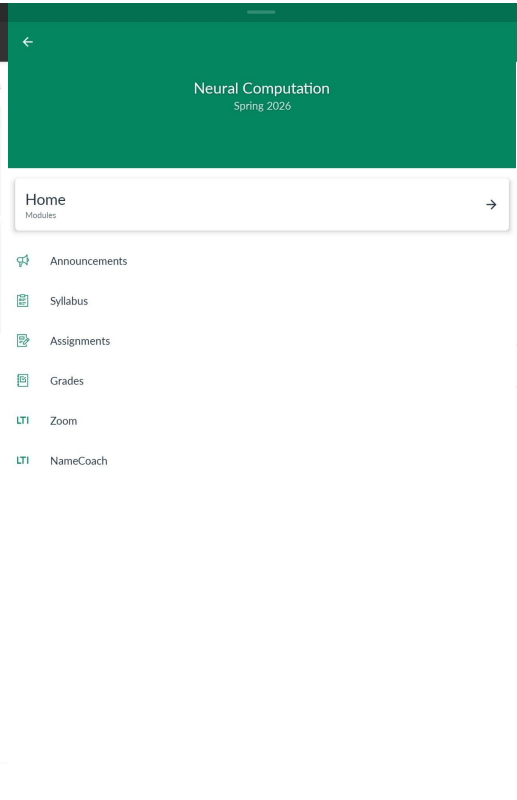
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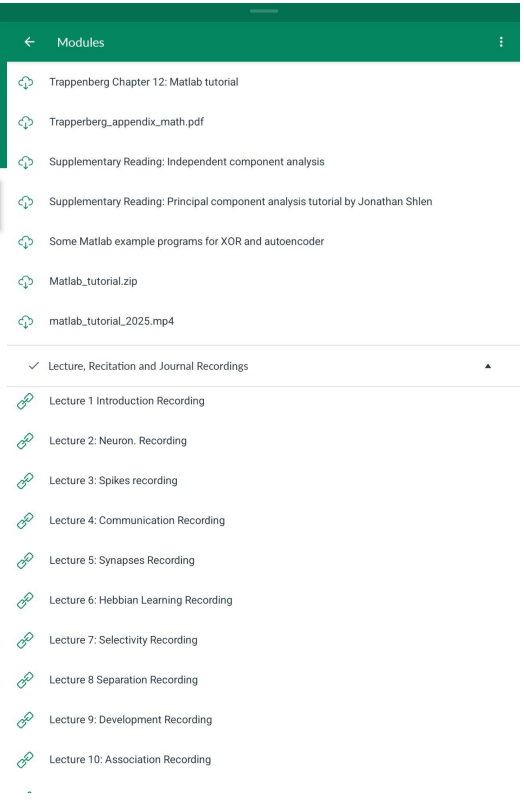
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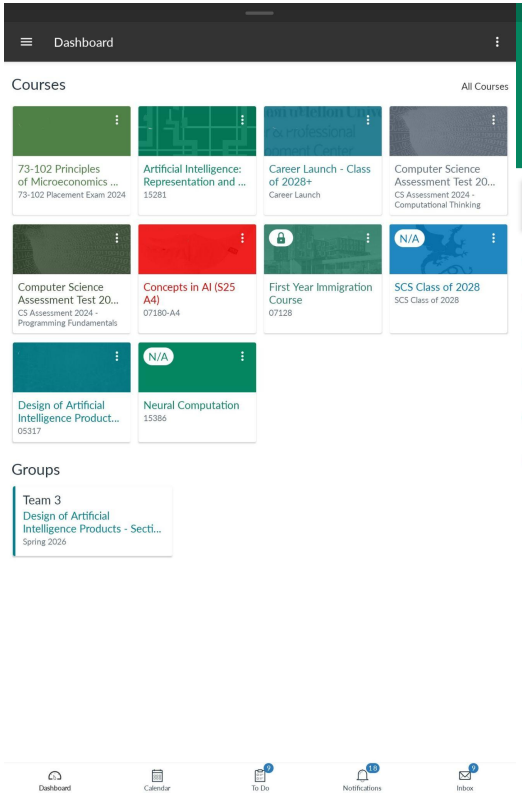
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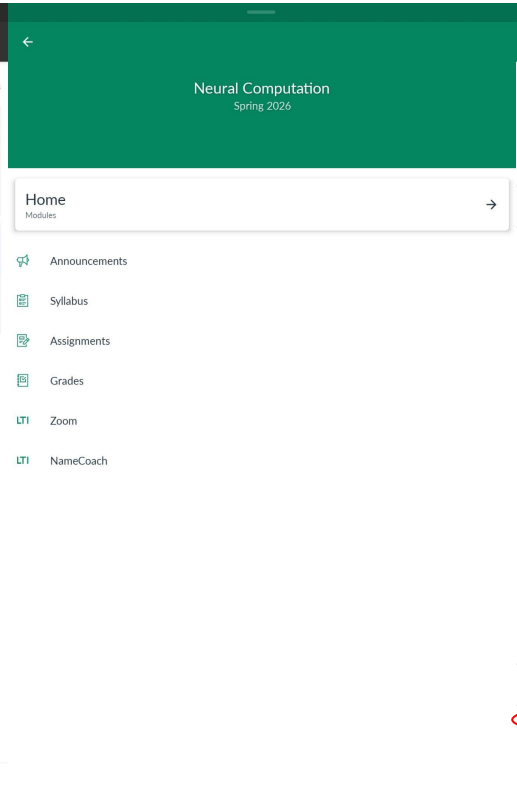
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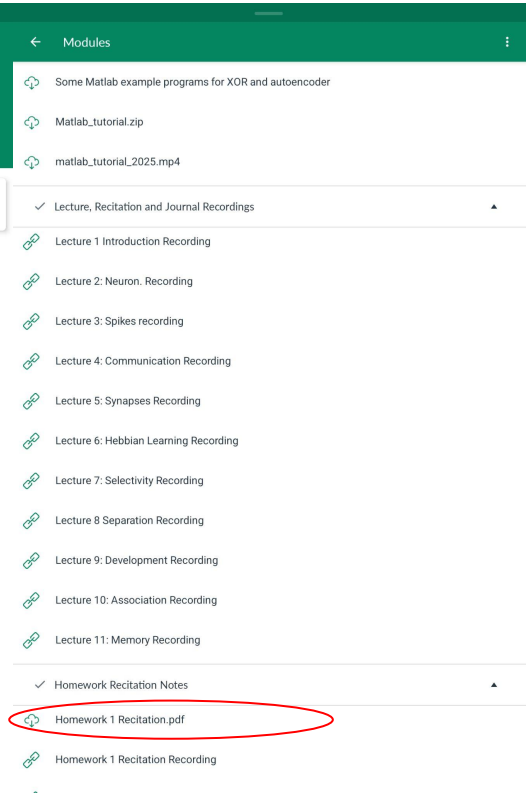
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Canvas Quick Click

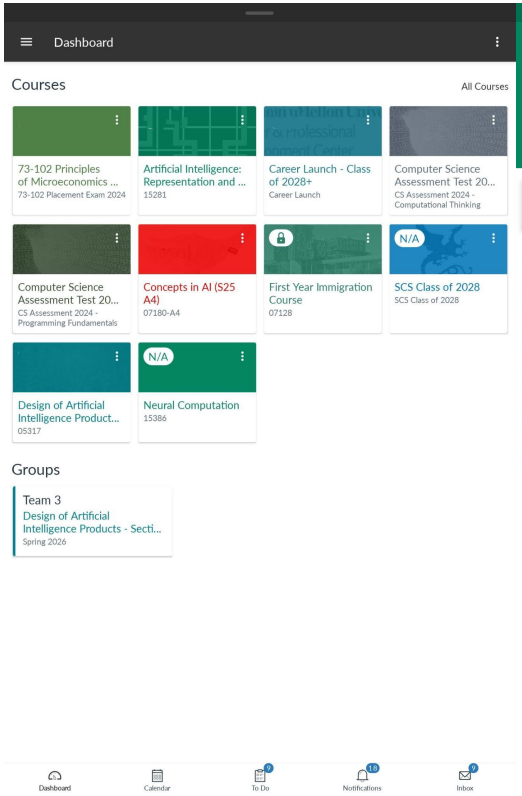
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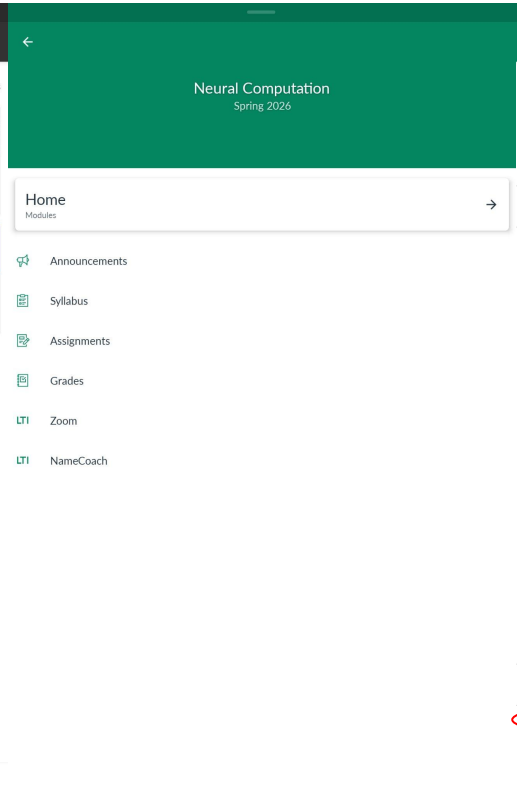
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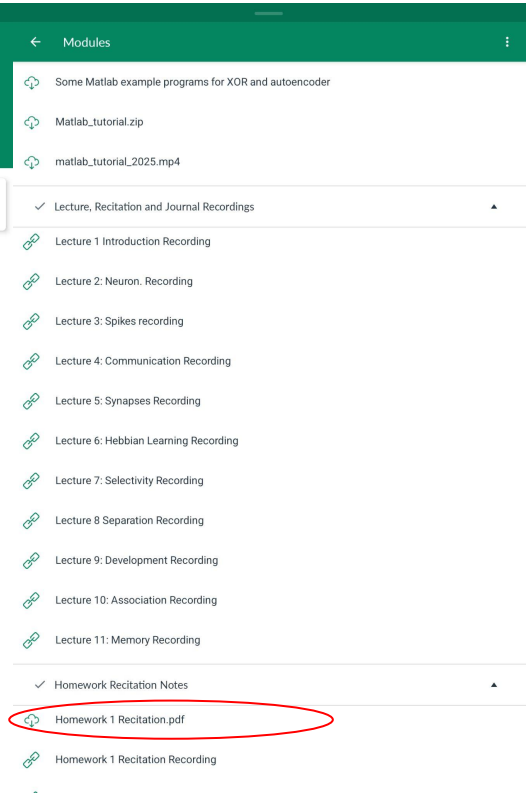
Co-creation of Value

Technical Feasibility

Financial feasibility







Canvas Quick Click

Problem

Solution

Co-creation of Value

Technical Feasibility

Financial feasibility

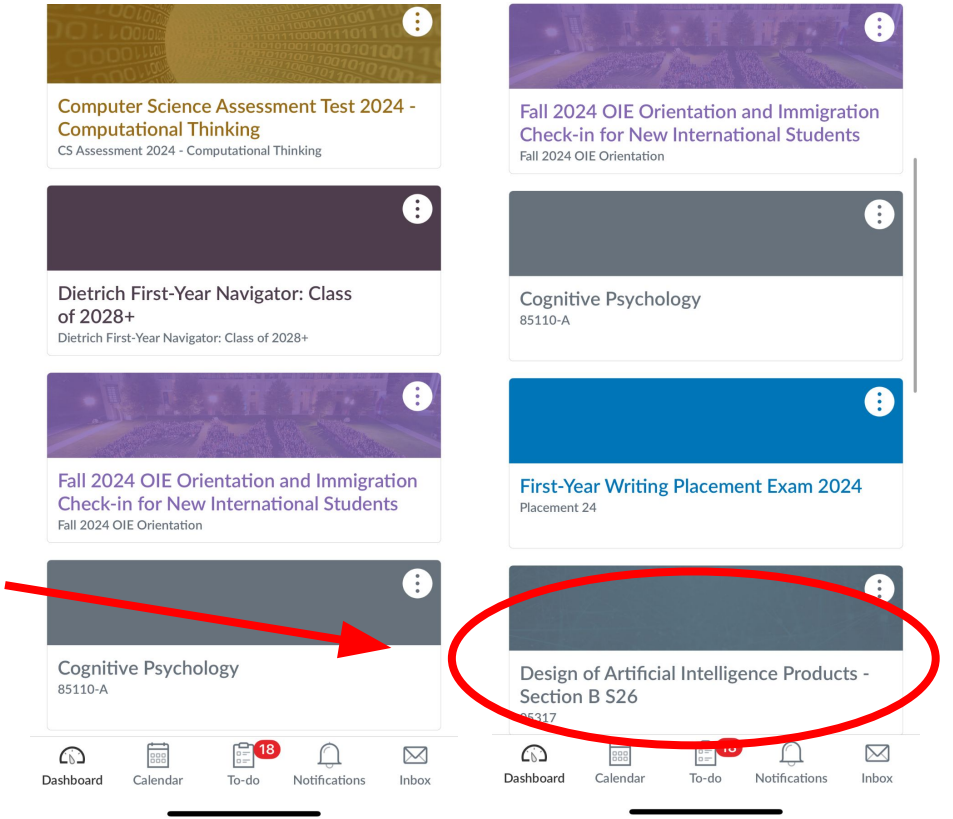
Current Non-adaptive Navigation

What the system does now:

- Static ordering
- Same structure for every user
- No memory of behavior

Frequently visited class not prioritized

User opens this class daily → still ranked 6th



Canvas Quick Click

Problem

Solution

Co-creation of Value

Technical Feasibility

Financial feasibility

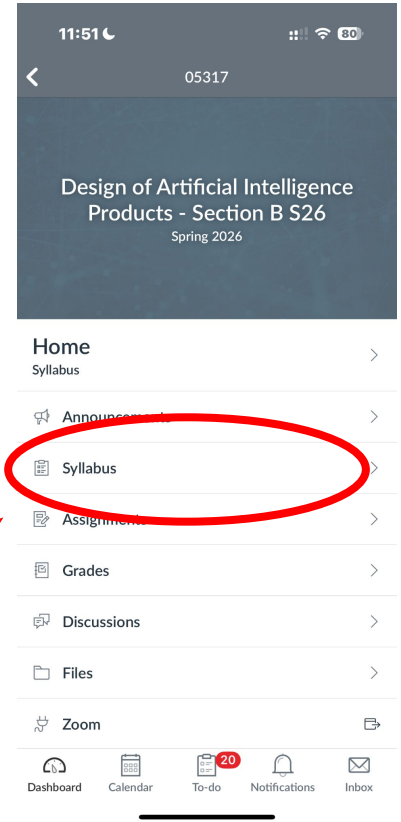
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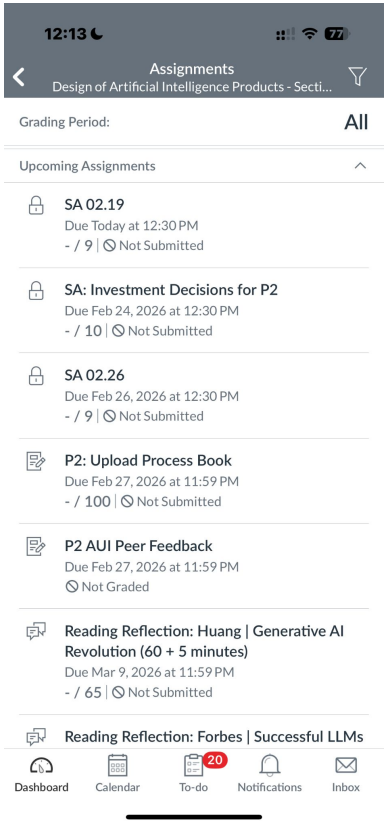
What the system does now:

- Static ordering
- Same structure for every user
- No memory of behavior

System adds friction to routine actions

Assignments accessed most → still nested equally





Canvas Quick Click

Problem

Solution

Co-creation of Value

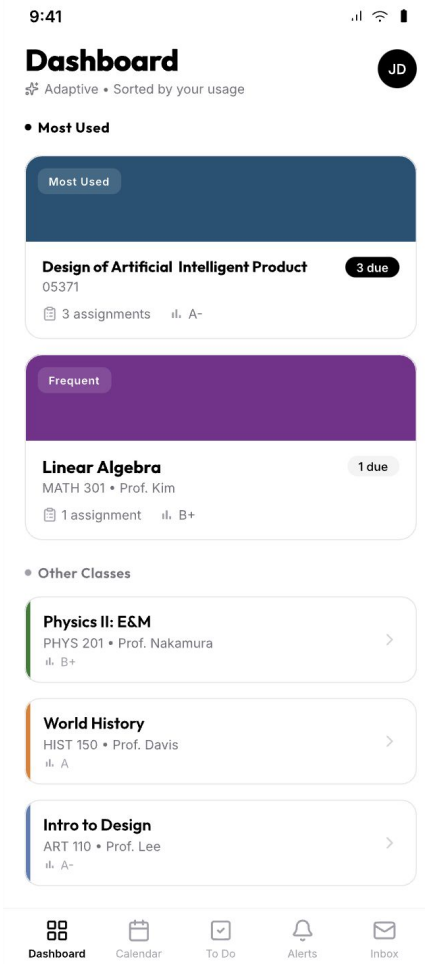
Technical Feasibility

Financial feasibility

Adaptive Navigation

Courses adaptively reorder on the dashboard

- Prioritize the most frequently used classes, based on user interaction/schedule patterns.
- Reduces navigation effort and improves task efficiency.



Canvas Quick Click

Problem

Solution

Co-creation of Value

Technical Feasibility

Financial feasibility

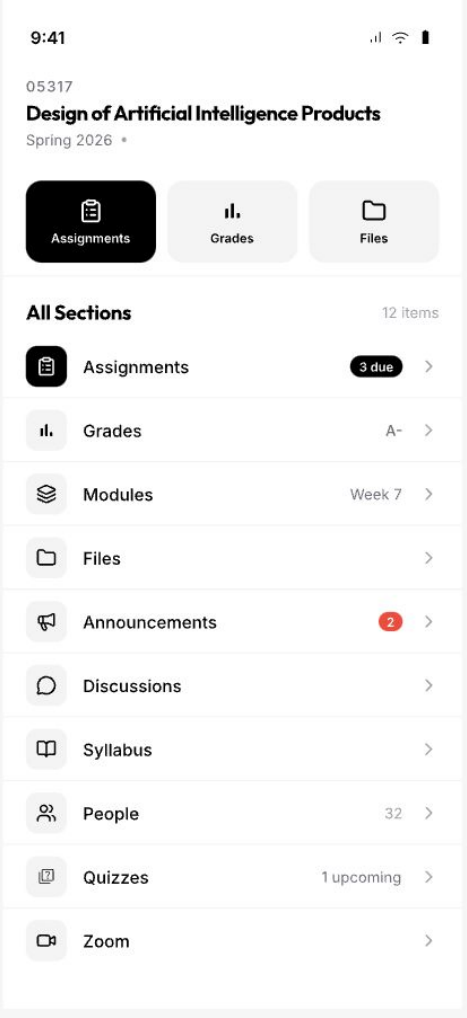
Adaptive Navigation

Organize Sections by Most Used

- Automatically surface most accessed sections at the top
- Reduce scrolling + navigation steps

Introduce horizontal tab bar

- Display the top 2–3 most accessed sections as tappable tabs
- Enable one-tap navigation to high-priority content



Co-Creation of Value



Students

- Reduces navigation friction (especially during urgent situations) and reduces cognitive load



Teachers

- Students engage with Canvas more, allowing teachers to manage and communicate with their students easier

Co-Creation of Value

Universities

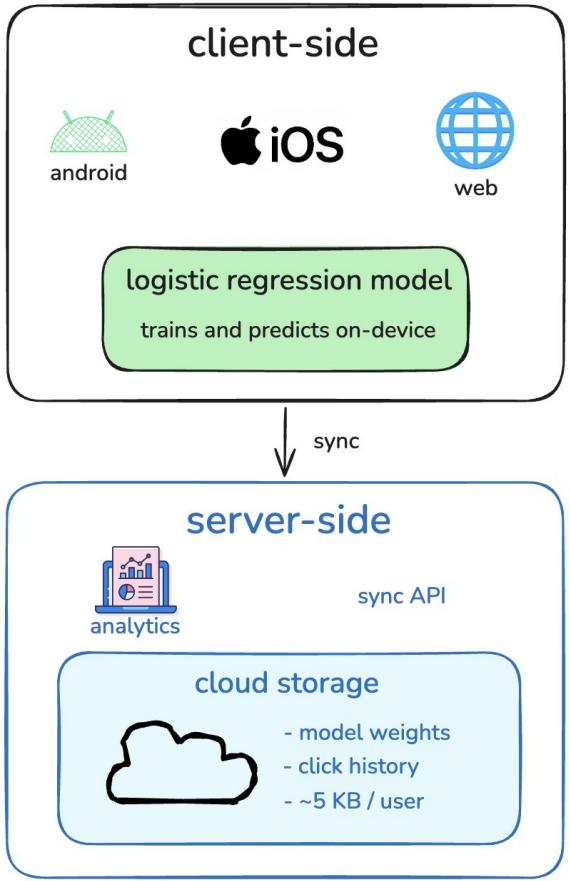
- More student successes
- Higher student satisfaction and institutional reputation

Canvas Business

- Gives data used to train AUI
  Provides data on high engagement and usage
- More subscriptions purchases, generating more revenue

Technical Feasibility

Simple Architecture

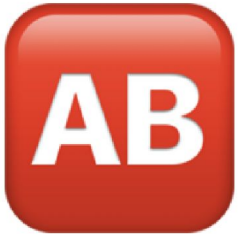


Cost Structure & ROI

Development



fullstack engineer

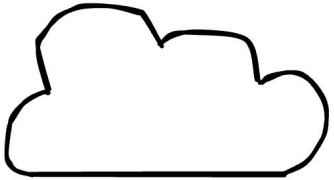


A/B testing

Operations



analytics



cloud storage

Financial Viability

Canvas contracts = ~**$10**/user.

$10/user * 10 new universities * 10,000 students **= $1,000,000**

Quick Click costs ~$50,000 to develop, **10-20x ROI**

User & Company Applicability

- All users will be able to access their work easier
- Canvas can collect data to market benefits to new organizations

Cost to Develop

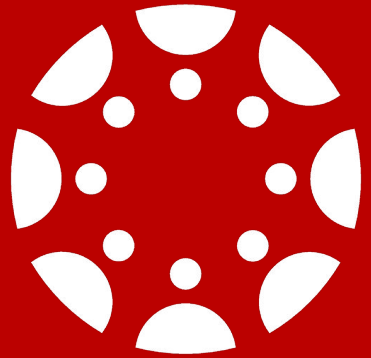
- Cheap to develop
  - Reorganizing already existing content
- Cost to maintain already present in Canvas

Customer Retention

- Current user experience will be improved

Thank You!

Questions?



Canvas Quick Click

Slide Title



Problem



Solution



Co-creation of Value



Technical Feasibility



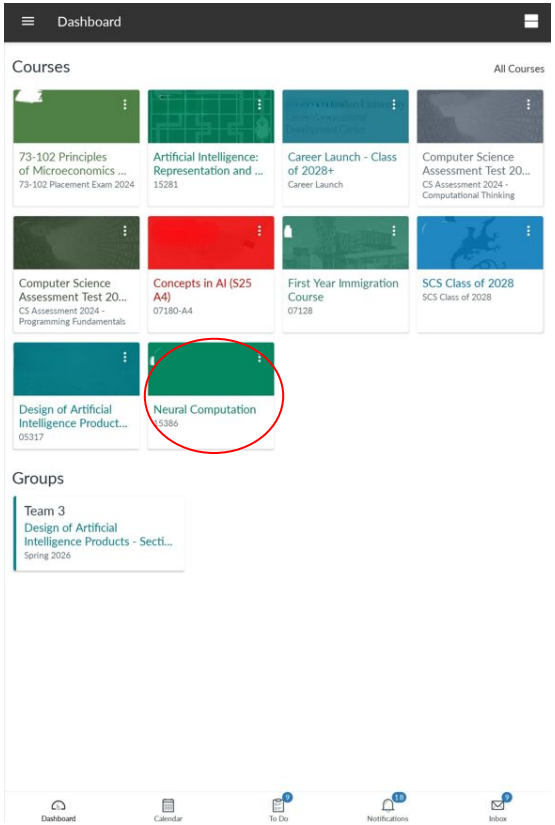
Financial feasibility

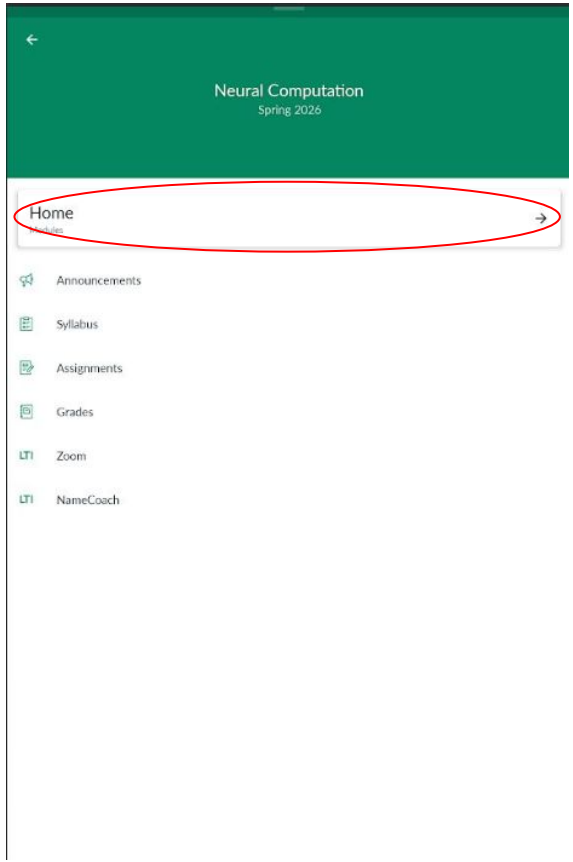
P2: Canvas AUI Draft Pitch

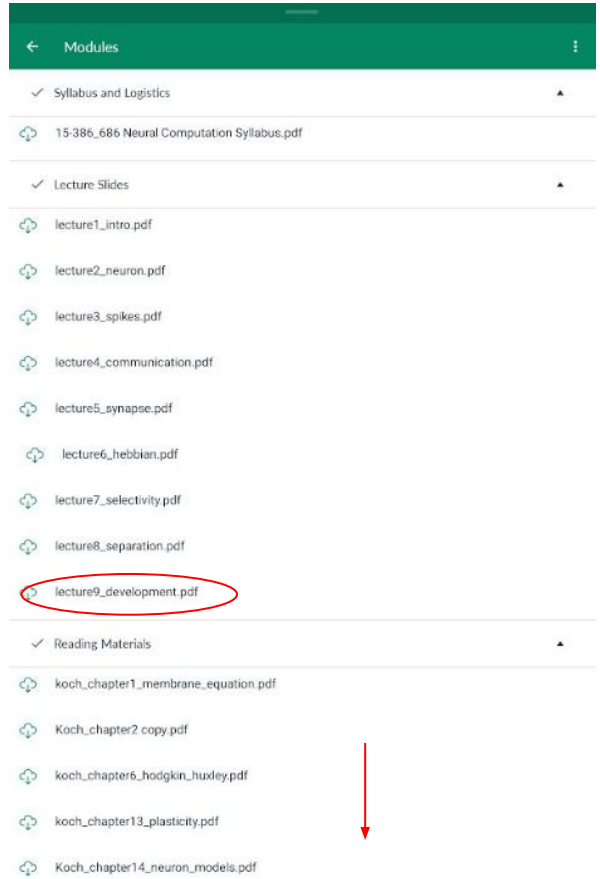
Team 3: Caroline Shi, Isabella Webber, Peter Shpelfogel, Priscilla Park, Dora Chen, Davis Germain

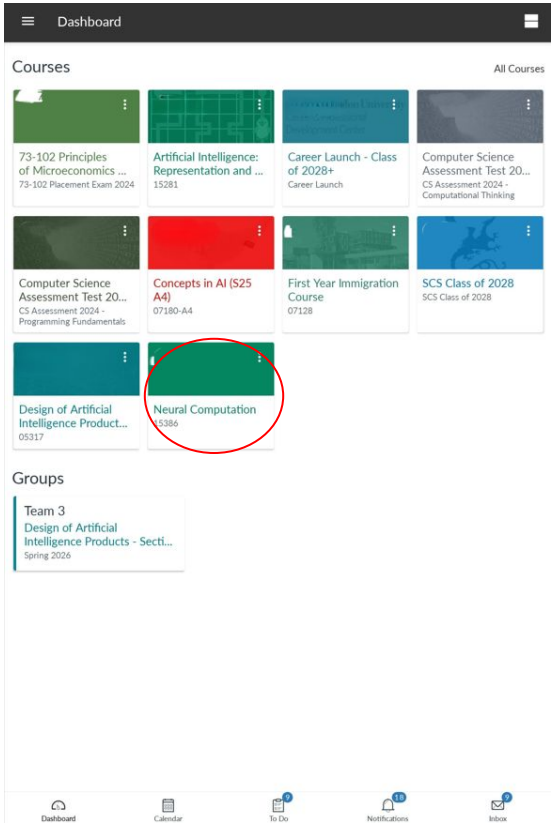
Tasks

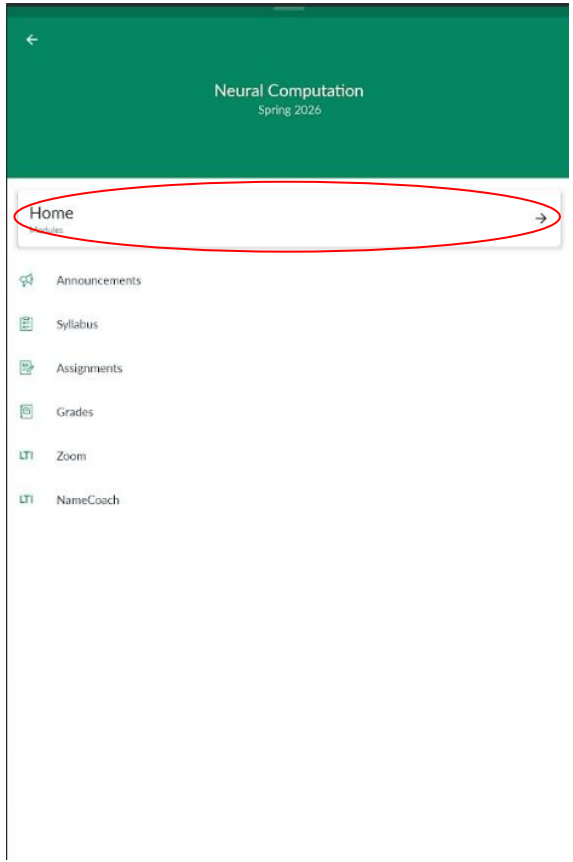
- Personal story (hook) - isabella
- Wireframe (not necessarily figma for 2/17 [template]) - peter, dora
  - BAD WIREFRAME
  - Important to mention on tablet/phone (all mobile)
  - Solution & GOOD WIREFRAME (+ user desirability)
- Co-creation of value (value flow model) - Priscilla Park
- Financial feasibility (talk about potential competition and how this beats out & we get user data which we can use for financial purposes) - davis
- Technical feasibility (user data: frequency, time-stamps) - Caroline

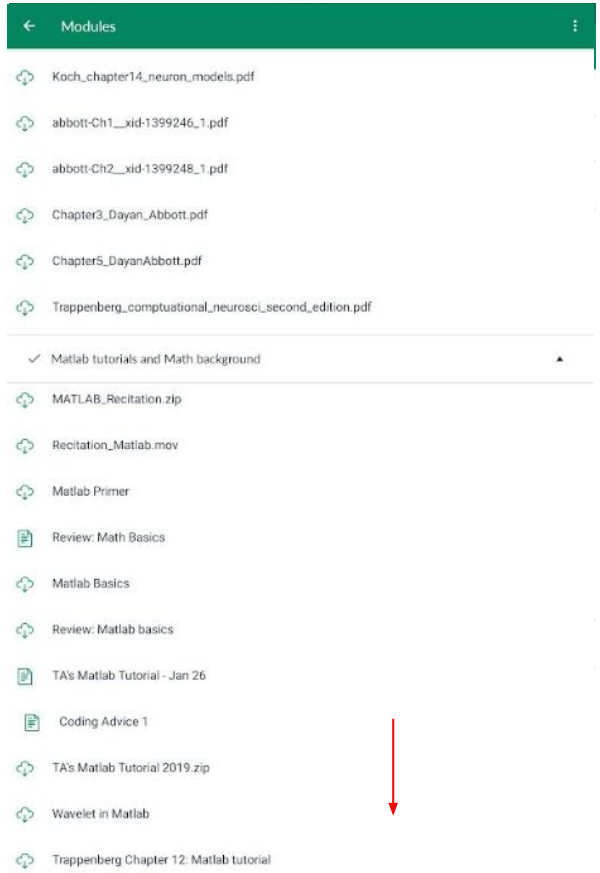

Dashboard
Courses
All Courses
73-102 Principles of Microeconomics ...
73-102 Placement Exam 2024
Artificial Intelligence: Representation and ...
15281
Career Launch - Class of 2028+
Career Launch
Computer Science Assessment Test 20...
CS Assessment 2024 - Computational Thinking
Computer Science Assessment Test 20...
CS Assessment 2024 - Programming Fundamentals
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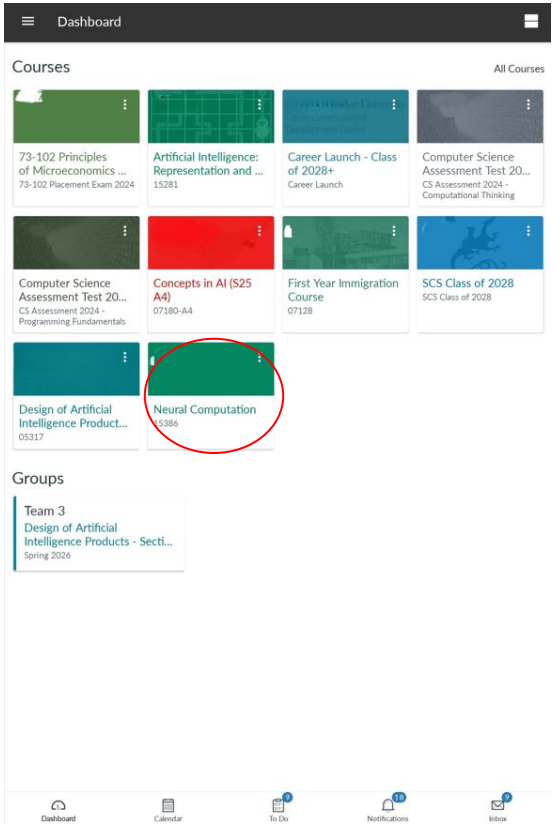

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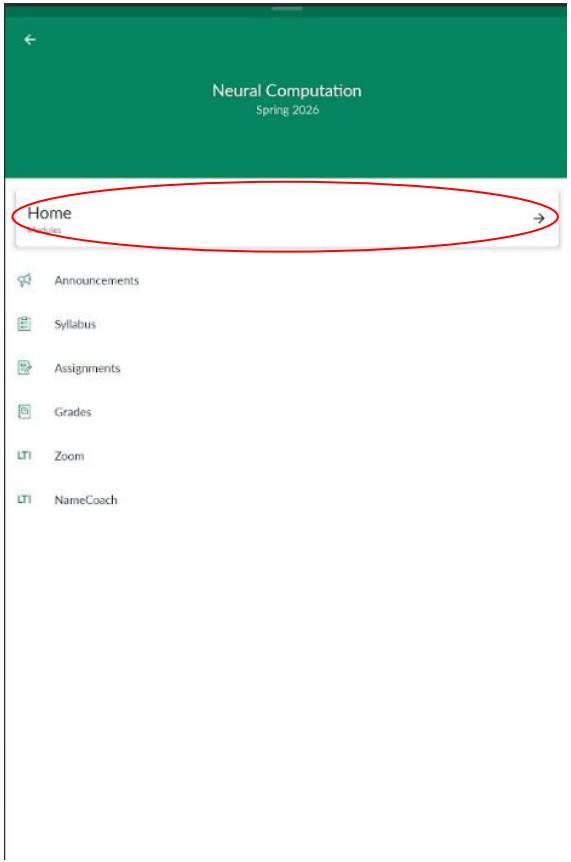

Modules
Syllabus and Logistics
15-386_686 Neural Computation Syllabus.pdf
Lecture Slides
lecture1_intro.pdf
lecture2_neuron.pdf
lecture3_spikes.pdf
lecture4_communication.pdf
lecture5_synapse.pdf
lecture6_hebbian.pdf
lecture7_selectivity.pdf
lecture8_separation.pdf
lecture9_development.pdf
Reading Materials
koch_chapter1_membrane_equation.pdf
Koch_chapter2 copy.pdf
koch_chapter6_hodgkin_huxley.pdf
koch_chapter13_plasticity.pdf
Koch_chapter14_neuron_models.pdf

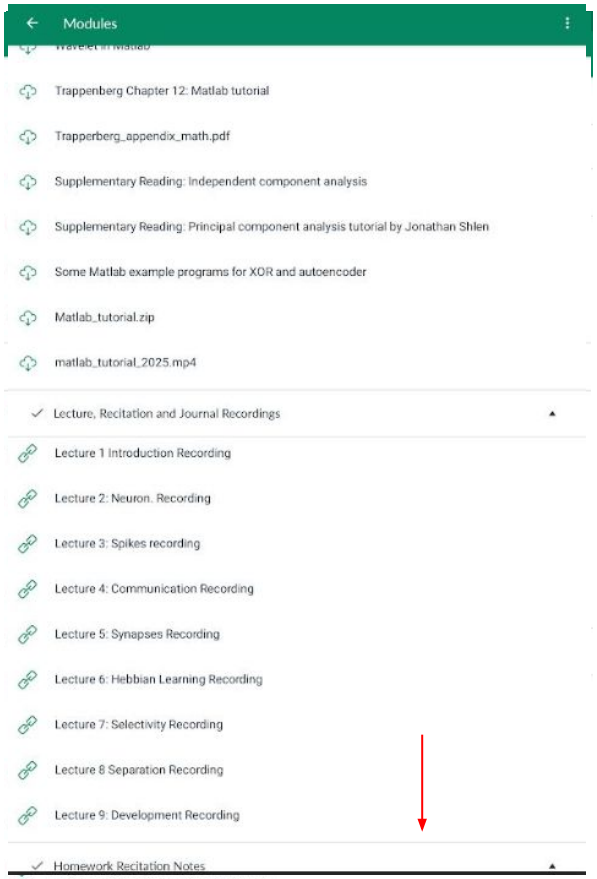

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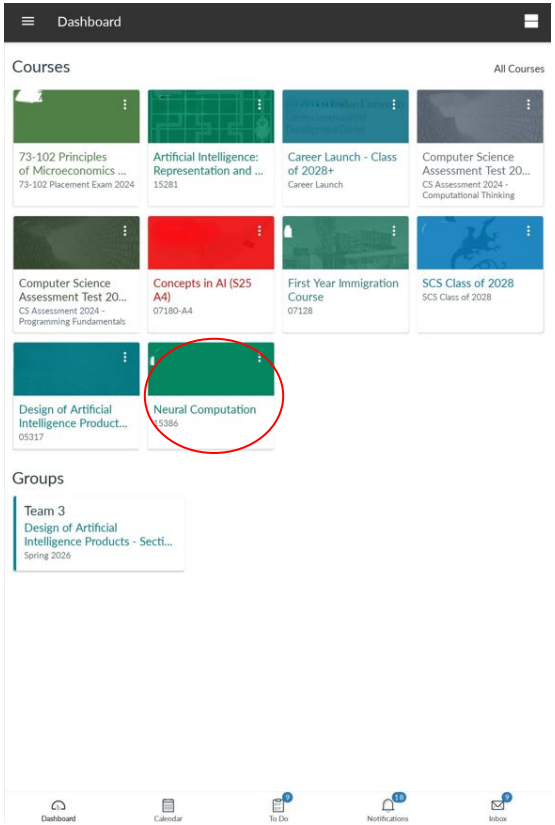

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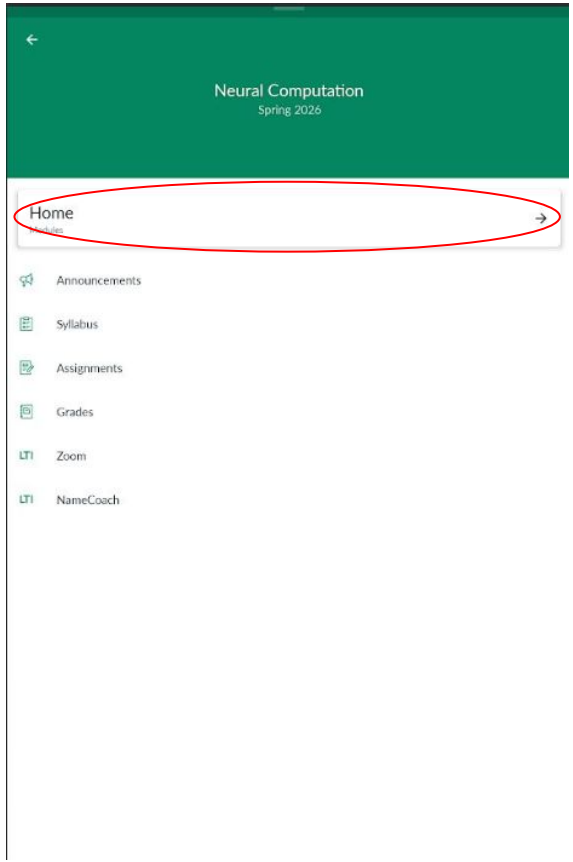

Modules
Koch_chapter14_neuron_models.pdf
abbott-Ch1__xid-1399246_1.pdf
abbott-Ch2__xid-1399248_1.pdf
Chapter3_Dayan_Abbott.pdf
Chapter5_DayanAbbott.pdf
Trappenberg_comptuational_neurosci_second_edition.pdf
Matlab tutorials and Math background
MATLAB_Recitation.zip
Recitation_Matlab.mov
Matlab Primer
Review: Math Basics
Matlab Basics
Review: Matlab basics
TA's Matlab Tutorial - Jan 26
Coding Advice 1
TA's Matlab Tutorial 2019.zip
Wavelet in Matlab
Trappenberg Chapter 12: Matlab tutorial

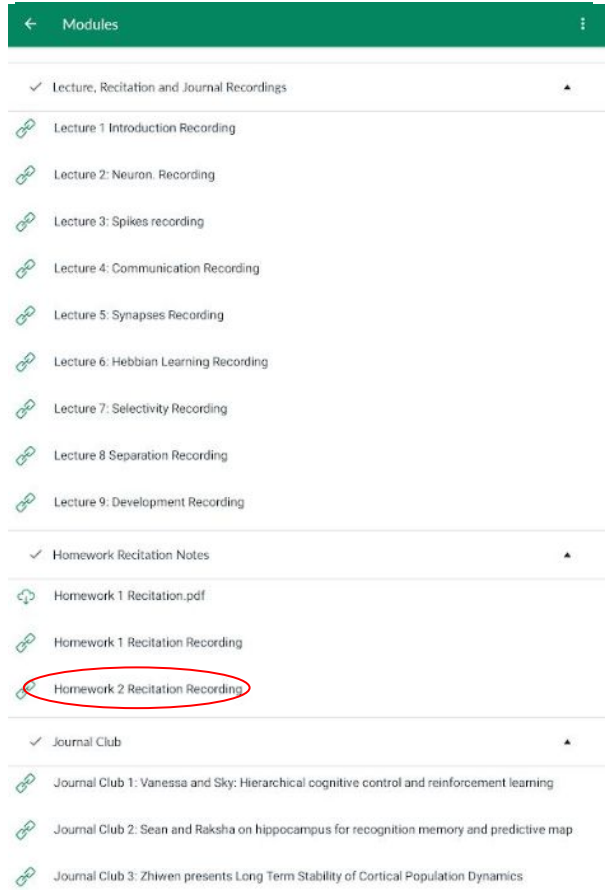

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Modules
Trappenberg Chapter 12: Matlab tutorial
Trapperberg_appendix_math.pdf
Supplementary Reading: Independent component analysis
Supplementary Reading: Principal component analysis tutorial by Jonathan Shlen
Some Matlab example programs for XOR and autoencoder
Matlab_tutorial.zip
matlab_tutorial_2025.mp4
Lecture, Recitation and Journal Recordings
Lecture 1 Introduction Recording
Lecture 2: Neuron. Recording
Lecture 3: Spikes recording
Lecture 4: Communication Recording
Lecture 5: Synapses Recording
Lecture 6: Hebbian Learning Recording
Lecture 7: Selectivity Recording
Lecture 8 Separation Recording
Lecture 9: Development Recording
Homework Recitation Notes


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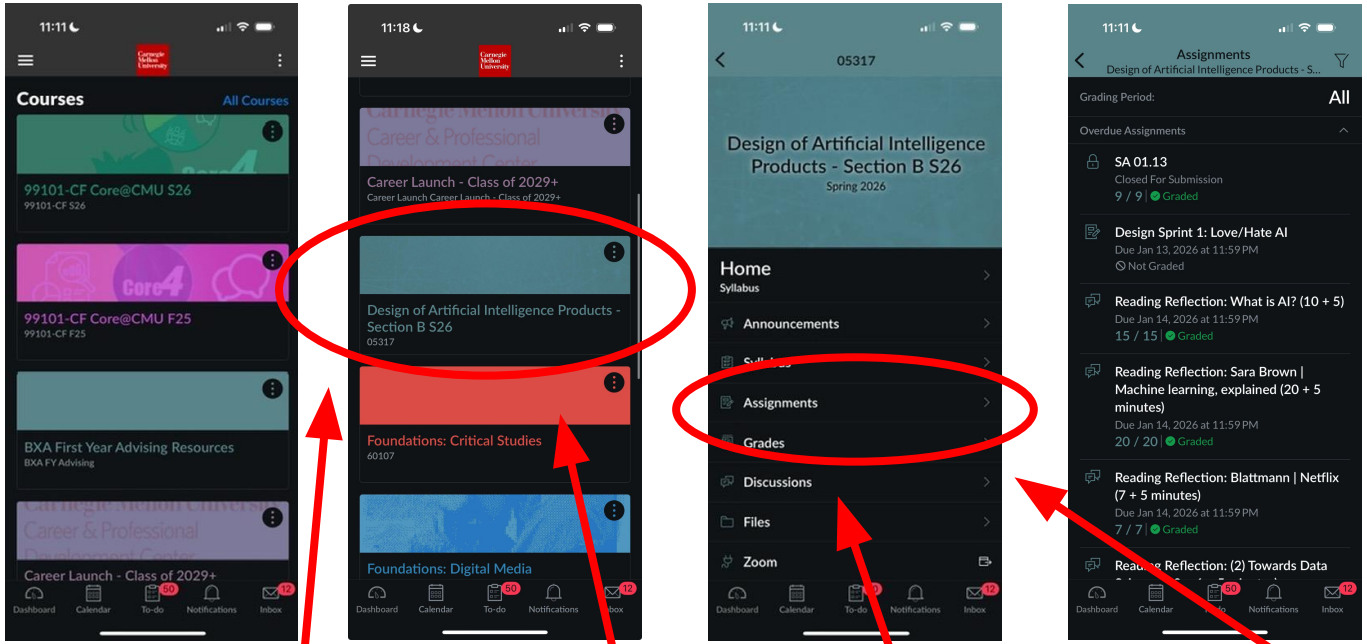

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Lecture 7: Selectivity Recording
Lecture 8 Separation Recording
Lecture 9: Development Recording
Homework Recitation Notes
Homework 1 Recitation.pdf
Homework 1 Recitation Recording
Homework 2 Recitation Recording
Journal Club
Journal Club 1: Vanessa and Sky: Hierarchical cognitive control and reinforcement learning
Journal Club 2: Sean and Raksha on hippocampus for recognition memory and predictive map
Journal Club 3: Zhiwen presents Long Term Stability of Cortical Population Dynamics

The Problem(s)

- Home page still shows earlier courses over frequently-accessed courses.
- The organization of Canvas pages relies entirely on how the teacher chooses to set up the course pages.
- As courses go on for longer, earlier lecture slides/assignments/handouts show up earlier in dropdowns (which display all items, rather than top 3 or 5).
- Even if students never use a certain tab or access a certain dropdown, the order on the homepage and Modules page is fixed and will not change over the semester.
- Difficult to locate needed materials for students, which is problematic for time-sensitive homework assignments and in-class assignments.

Current Non-adaptive Navigation



Frequently visited class not prioritized

User opens this class daily → still ranked 5th

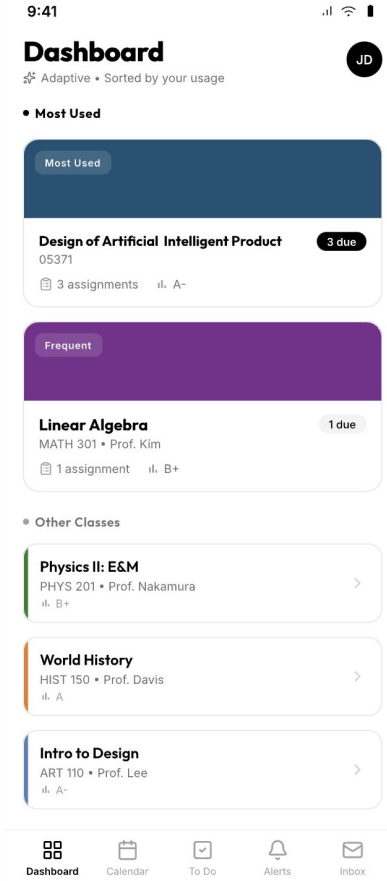
System adds friction to routine actions

Assignments accessed most → still nested equally

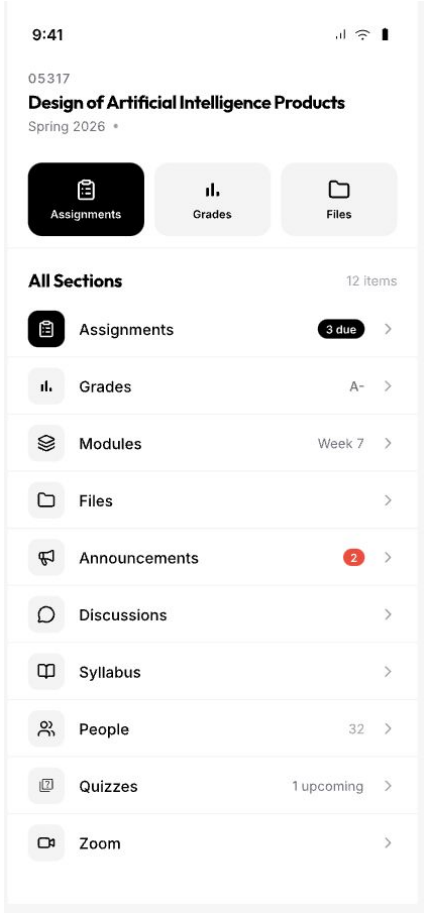
What the system does now:

- Static ordering
- Same structure for every user
- No memory of behavior

Adaptive Solution



- Courses adaptively reorder to prioritize most frequently used classes/based on the schedule
- Reduces navigation effort and improves task efficiency.



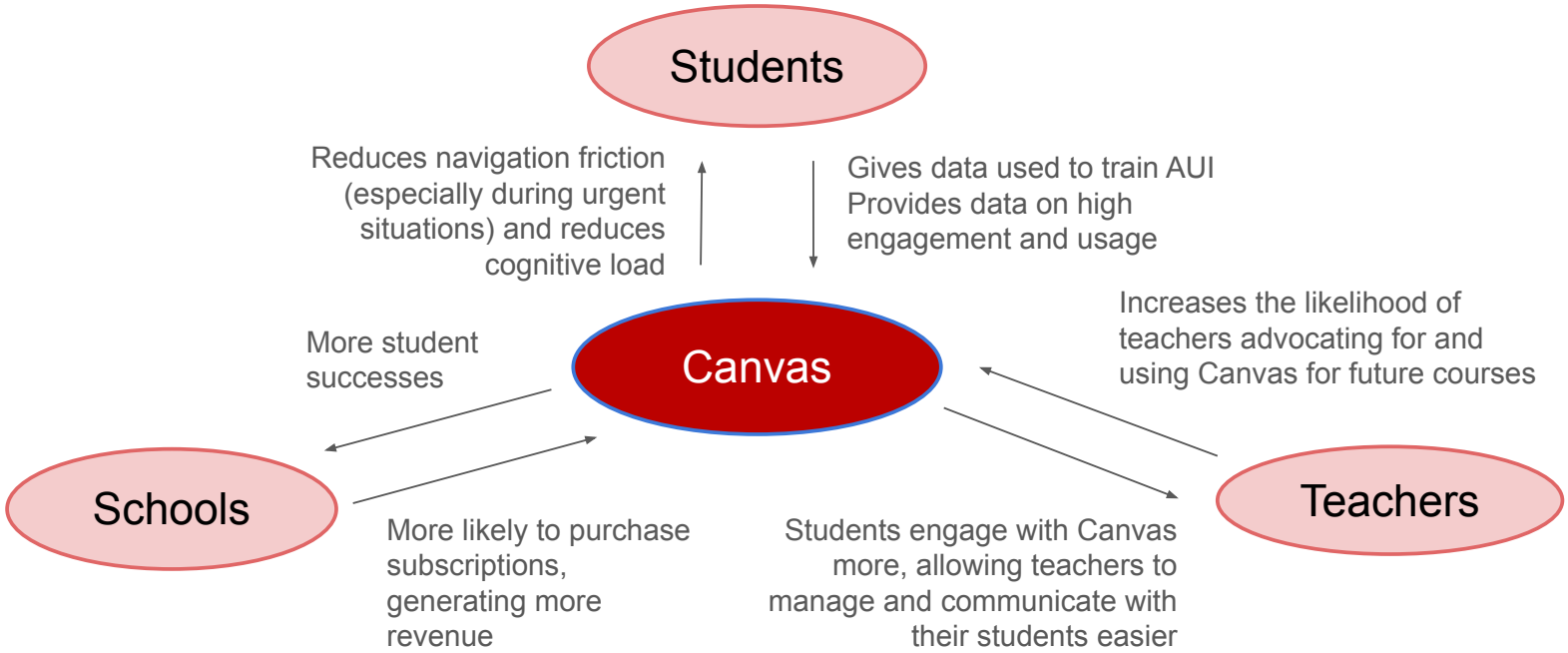
Organize Sections by Most Used

- Automatically surface most accessed sections at the top
- Preserve full section list while adaptively prioritizing high-usage items
- Reduce scrolling + navigation steps

Introduce horizontal tab bar below each class header that highlights highly used sections.

- Display the top 2–3 most accessed sections as tappable tabs
- Enable one-tap navigation to high-priority content

Co-creation of value (Value Flow Model)



Technical Feasibility

Simple Architecture

- Individual user model (no global complexity)
- Simple logistic regression model, allows for on-device training
    - Features: frequency, recency, time-of-day, day-of-week
- Lightweight cloud storage of click history
- Model trains client-side, syncs weights to cloud

Cost Structure & ROI

Development

- Full-stack engineer
- A/B test framework

Operations

- Cloud storage
- Analytics

Financial Feasibility

User & Company Applicability

- All users will be able to access their work easier
- Canvas can collect data to observe benefits of the new design
    - Market to new customers against competitors to show benefits

Cost to Develop

- Cheap to develop
    - Removing already existing content
- Cost to maintain was already present in the product

Customer Retention

- Current customers' experience will be improved
- Will reduce layers of friction between users and their work
    - Attracts new customers